

Brain MRI

ACG: A-0047 (AC)

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Clinical Indications

- Brain MRI may be indicated for **1 or more** of the following(1)(2)(3):
 - ☐ Acoustic neuroma (vestibular schwannoma), as indicated by **1 or more** of the following(20)(21):[N](#)
 - Monitoring of acoustic neuroma (vestibular schwannoma) without surgery; intervals include **1 or more** of the following:
 - Six and twelve months after initial diagnosis
 - Annually for next 5 years
 - Every 1 to 2 years thereafter
 - Monitoring after radiosurgical removal of acoustic neuroma (vestibular schwannoma): every 3 years
 - ☐ Anatomy or structural defect evaluation needed, as indicated by **1 or more** of the following(24):[N](#)
 - Chiari malformation(26)
 - Congenital anomaly, suspected (eg, Dandy-Walker malformation, septo-optic dysplasia, agenesis of corpus callosum)(25)(27)(28)(29)(30)
 - Horner syndrome, and central nervous system cause suspected(31)
 - Hydrocephalus, as indicated by **1 or more** of the following(32)(33)(34):
 - Adult, child, or infant with known hydrocephalus, and need for follow-up
 - Adult with suspected normal pressure hydrocephalus, as indicated by **1 or more** of the following(35)(36)(37)(38):
 - Gait disturbance
 - Recent onset of dementia or worsening of established dementia
 - Urinary incontinence
 - Child or infant with suspected hydrocephalus, as indicated by **1 or more** of the following(39)(40):
 - Abnormal findings on neurologic examination (eg, papilledema, gait disturbance, increased deep tendon reflexes, spasticity, clonus, Babinski sign)
 - Downward gaze (ie, sunset gaze)
 - Increasing head size beyond normal rate
 - Persistence of posterior fontanelle beyond 6 to 8 weeks of age
 - Signs or symptoms suggestive of hydrocephalus (eg, irritability, lethargy, decreased mentation, vomiting, headache)
 - Widening or bulging anterior fontanelle, or persistence of anterior fontanelle beyond 18 months of age
 - Ventriculoperitoneal shunt in place and need for further evaluation, as indicated by **1 or more** of the following(41):
 - Accelerated head growth

- Acute neurologic deficit
 - Decreased mentation
 - Headache
 - Irritability
 - Lethargy
 - Loss of developmental milestones
 - New or more frequent seizures
 - Shunt reservoir malfunctioning
 - Unexplained fever
 - Vomiting
- Ventriculoperitoneal shunt in place and need for follow-up(38)
- Neurocutaneous disorders (eg, hereditary hemorrhagic telangiectasia, neurofibromatosis, Sturge-Weber syndrome, tuberous sclerosis complex, von Hippel-Lindau disease)(42)(43)(44)(45)(46)
- ☐ Anosmia, and suspicion for central nervous system etiology, as indicated by **1 or more** of the following(47)(48)(49):[N](#)
 - Anosmia present since birth(52)
 - Cognitive impairment
 - History of head or facial trauma(53)
 - Hypogonadism(52)
 - Olfactory hallucinations(52)
 - Parkinson disease or other neurodegenerative disorder, known or suspected(50)(51)
- ☐ Cancer or neoplasm evaluation, staging, or surveillance needed, as indicated by **1 or more** of the following(54)(55)(56)(57)(58)(59):[N](#)
 - Abnormal or nondiagnostic CT scan finding, and need for further evaluation
 - Bone neoplasm (benign or malignant) involving calvaria or skull base, known or suspected (eg, chondrosarcoma, chordoma, Ewing sarcoma family of tumors, giant cell tumor of bone, osteosarcoma, metastatic tumor), as indicated by **1 or more** of the following(68)(69)(70):
 - Initial evaluation or staging(71)(72)
 - Monitoring disease progression or response to treatment
 - Post-treatment surveillance
 - Focal neurologic signs or symptoms in cancer patient(73)(74)(75)(76)(77)
 - Germ cell tumor, gonadal (eg, testicular seminoma or nonseminomatous germ cell tumor) or extragonadal (eg, retroperitoneal or mediastinal seminoma or nonseminomatous germ cell tumor), as indicated by **1 or more** of the following(78)(79)(80)(81):
 - Initial staging or risk stratification in patient with high-risk disease^[A](82)
 - Evaluation of suspected recurrence (eg, due to elevated or rising tumor marker levels)
 - Histiocytic or dendritic cell neoplasm^[B] with involvement of central nervous system, skull, or craniofacial bones, known or suspected(83)(86)(87)
 - Intraocular or orbital tumor, known or suspected(70)(88)(89)(90)(91)(92)
 - Lymphoma, known or suspected, as indicated by **1 or more** of the following(93)(94)(95):
 - High-risk subtype (eg, diffuse large B-cell lymphoma, adult T-cell lymphoma), and need for initial evaluation, staging, or monitoring response to treatment(96)(97)
 - Known central nervous system involvement, and need for monitoring response to treatment
 - Suspected central nervous system involvement (eg, due to neurologic signs or symptoms), and need for evaluation, staging, or treatment planning
 - Melanoma, as indicated by **1 or more** of the following(98):
 - Initial staging of metastatic disease
 - Post-treatment surveillance of stage IIB to stage IV disease
 - Meningioma (benign, atypical, or unresected) managed by observation only: at 3, 6, and 12 months, then every 6 to 12 months for 5 years, then every 1 to 3 years as indicated
 - Neuroendocrine tumor, poorly differentiated(99)(100)
 - Non-small cell lung cancer, as indicated by **1 or more** of the following(101):
 - Initial staging needed for stage IB disease in high-risk patient (eg, central tumor location)
 - Initial staging needed for stage II disease or greater
 - Locoregional recurrence and need to assess for brain metastases(102)
 - Pituitary or sellar tumor, known or suspected, as indicated by **1 or more** of the following(65)(103)(104)(105)(106):
 - Pituitary tumor signs or symptoms, as indicated by **1 or more** of the following(107)(108)(109):
 - Abnormal sella turcica on plain x-ray or CT scan
 - Acromegaly
 - Amenorrhea or oligomenorrhea in premenopausal female
 - Central cortisol insufficiency
 - Central hyperthyroidism
 - Central hypothyroidism

- Cushing disease
 - Diabetes insipidus
 - Galactorrhea
 - Growth hormone deficiency(110)
 - Hirsutism
 - Hyperprolactinemia(111)(112)
 - Hypogonadotropic hypogonadism
 - Oculomotor palsies(113)
 - Unexplained visual field deficit(113)(114)(115)
- Post-treatment surveillance of pituitary tumor(65)(107)
- Surveillance of pituitary incidentaloma; intervals include **1 or more** of the following(67):
 - For macroincidentaloma: 6 months after initial scan, then annually for next 3 years if stable in size, and less often thereafter[C]
 - For microincidentaloma: 1 year after initial scan, then every 1 to 2 years for next 3 years if stable in size, and less often thereafter[D]
- Planning for interventional therapy (eg, stereotactic radiosurgery)(116)(117)(118)
- Primary central nervous system tumor (eg, adult-type or pediatric-type diffuse glioma, astrocytic glioma, glioneuronal tumor, ependymoma, medulloblastoma, other central nervous system embryonal tumor, meningioma, primary central nervous system lymphoma, central nervous system germ cell tumor), and need for **1 or more** of the following(73)(119)(120)(121)(122)(123)(124):
 - Initial evaluation or staging(125)(126)
 - Monitoring disease progression or response to treatment
 - Post-treatment surveillance
- Primary non-central nervous system tumor, and need for diagnosis, staging, or monitoring of suspected brain or leptomeningeal metastasis, as indicated by **1 or more** of the following(76)(127):
 - Known metastatic disease, and need for monitoring disease progression or response to treatment
 - Surveillance for recurrence of treated brain metastasis every 2 to 3 months for 1 to 2 years, then every 4 to 6 months indefinitely(73)
 - Suspected metastatic disease (eg, due to neurologic signs or symptoms in patient with cancer), and need for evaluation, staging, or treatment planning
- Screening for central nervous system tumor in patient with high-risk genetic disorder or syndrome (eg, Li-Fraumeni syndrome, multiple endocrine neoplasia type 1, tuberous sclerosis complex, von Hippel-Lindau syndrome)(43)(128)(129)(130)(131)(132)(133)
- Small cell lung cancer, and need for **1 or more** of the following(102)(134)(135)(136):
 - Initial staging
 - Monitoring treatment response
 - Recurrence or progression, known or suspected
 - Surveillance of patient without brain metastases after primary treatment: every 3 to 4 months for first year, then every 6 months
 - Surveillance of patient with asymptomatic brain metastases who is receiving systemic therapy before brain radiotherapy: after every 2 cycles of treatment and at completion of therapy
- Cerebral edema, suspected(137)(138)[N]
- Cerebrospinal fluid leak, suspected(139)(140)(141)(142)[N]
- Delirium or change in level of consciousness(143)(144)[N]
- ☐ Dementia or mild cognitive impairment, known or suspected, and **1 or more** of the following(146)(147)(148)(149)(150):[N]
 - Initial diagnostic evaluation(168)(169)(170)(171)
 - Amyloid-related imaging abnormalities (ARIA), known, and need for evaluation for continued treatment with monoclonal antibody therapy for Alzheimer disease(167)(172)
 - Evaluation for ARIA prior to initiation or monitoring during treatment with monoclonal antibody therapy for Alzheimer disease(167)(172)
- ☐ Demyelinating disease, known or suspected (eg, multiple sclerosis), as indicated by **1 or more** of the following(173)(174)(175)(176)(177):[N]
 - Clinically isolated syndrome or signs and symptoms that cannot be otherwise explained (eg, temporary loss of vision, diplopia, pupillary dysfunction, ascending numbness or tingling, episodic clumsiness)(115)(185)(186)(187)
 - Multiple sclerosis, and need for further evaluation, as indicated by **1 or more** of the following(188):
 - Increasing cognitive impairment
 - Symptoms suggesting treatment failure
 - Transition from one subtype of multiple sclerosis to another is suspected (eg, from relapsing-remitting to secondary-progressive).(189)
 - Repeat study if diagnosis unclear after initial central nervous system imaging and after at least 90 days from onset of symptoms(178)(180)(190)

- Repeat study to monitor response to pharmacotherapy for multiple sclerosis; intervals include **1 or more** of the following(178)(180)(188):
 - Annually in stable patient
 - After initiation of treatment with new pharmacotherapy: 3 to 6 months
 - Patient at increased risk of complications from pharmacotherapy (ie, progressive multifocal leukoencephalopathy, inflammatory demyelination): 3 to 4 months
- Developmental delay(191)(192)(193):[N](#)
- ☐ Dizziness or vertigo, as indicated by **1 or more** of the following(195)(196)(197):[N](#)
 - Brainstem findings (eg, dysarthria, Horner syndrome)
 - Cerebellar findings (eg, incoordination of voluntary movements, intention tremor, disorder of equilibrium or gait, diminished muscle tone)
 - Focal neurologic findings (eg, weakness, numbness, paresthesias on one side of body)
 - New-onset headache or neck pain(199)
 - Unresponsive to antibiotic treatment for sinus or ear infection
 - Unresponsive to symptomatic treatment
- ☐ Epilepsy or seizure disorder, suspected or known, as indicated by **1 or more** of the following(200)(201)(202)(203)(204):[N](#)
 - Abnormal neurologic examination (eg, focal deficit, stigmata of neurocutaneous or cerebral malformation, developmental delay or regression in child)
 - Change in seizure pattern
 - First focal seizure(205)
 - First unprovoked generalized seizure(209)(210)
 - More frequent seizures despite anticonvulsant medication(211)
 - Preoperative evaluation when surgery being considered(206)(207)(212)(213)
 - Seizure in child younger than 2 years, excluding those with febrile seizures
 - Status epilepticus(208)(214)
- ☐ Headache with possible underlying structural cause, as indicated by **1 or more** of the following(199)(215)(216)(217)(218):[N](#)
 - Personal history suggesting underlying infectious, inflammatory, or structural cause, as indicated by **1 or more** of the following:
 - Onset of headache before age 6 years or after age 50 years
 - Patient with history of cancer
 - Patient with HIV or immunosuppression
 - Signs or symptoms suggesting underlying infectious, inflammatory, or structural cause, as indicated by **1 or more** of the following:
 - Abnormal findings on neurologic examination, including altered mental status or personality(219)
 - Accompanied by seizure(219)
 - Accompanied by vomiting
 - Change in frequency, severity, or clinical features of headache from what patient has commonly experienced
 - Cluster-type headache(220)
 - Meningeal signs
 - Motor or sensory aura, or aura that has changed character
 - New or progressive headache that persists for days
 - Persistent headache without family history
 - Precipitated by exertion, coughing, sneezing, bending down, or sexual intercourse
 - Repeatedly awakens child from sleep or is present upon awakening
 - Systemic symptoms (ie, fever, myalgias, weight loss, scalp tenderness)
 - Temporal arteritis, suspected
 - Unresponsive to medical treatment
 - Worst headache of patient's life (ie, "thunderclap" headache)(219)(221)
- ☐ Hearing loss, as indicated by **1 or more** of the following(195)(196):[N](#)
 - Associated with neurologic symptoms or focal findings
 - Congenital sensorineural hearing loss
 - Postoperative assessment of acoustic neuroma (vestibular schwannoma) (eg, after resection or stereotactic radiosurgery) (20)
 - Preoperative imaging for cochlear implant
 - Unexplained unilateral sensorineural hearing loss documented by audiometry(222)
- ☐ Infection, known or suspected, as indicated by **1 or more** of the following(223)(224)(225)(226)(227):[N](#)
 - Abscess of brain, known, and need for follow-up(232)(233)
 - Abscess of brain, suspected (eg, history of sinus infection, headache, fever, neurologic abnormalities)(232)(233)
 - Immunosuppressed patient (eg, receiving chemotherapy, HIV-positive) with recent onset of neurologic abnormalities(234)(235)(236)(237)(238)
 - Meningitis, suspected, and need to confirm diagnosis or exclude contraindications to lumbar puncture(225)(231)(239)(240)

- Neurocysticercosis, suspected(241)
- Viral or bacterial encephalitis, suspected (eg, herpes infection, confusion, neurologic abnormalities)(242)(243)(244)(245)(246)
- Intracranial vasculitis, suspected, and need for evaluation of possible sequelae(247)(248)(249)(250)(251)(252)[N](#)
- Neonatal hypoxic-ischemic encephalopathy during first 7 days of life(253)(254)[N](#)
- ☐ Neurologic disease signs or symptoms, as indicated by **1 or more** of the following(49)(255)(256)(257):[N](#)
 - Ataxia or gait disturbance(198)
 - Change in speech or language (eg, dysarthria, aphasia)
 - Cranial nerve palsy(113)(258)
 - Focal sensory deficit (eg, numbness, paresthesias) of face, limb, or whole side of body
 - Focal weakness of face, limb, or whole side of body
 - Horner syndrome (unilateral miosis, ptosis, facial anhidrosis)(259)
 - Papilledema(113)
 - Visual disturbance (eg, diplopia, visual field defect, nystagmus, visual loss)(106)(113)
- ☐ Parkinson disease or other neurodegenerative disorders, as indicated by **1 or more** of the following(260)(261)(262)(263):[N](#)
 - Established Parkinson disease and **1 or more** of the following:
 - Atypical features unresponsive to levodopa
 - Prior to surgery or deep brain stimulation
 - New onset of symptoms suggestive of Parkinson disease or other neurodegenerative disorder(150)(165)(269)(270)(271)
- ☐ Precocious puberty (central), as indicated by **ALL** of the following(65)(272)(273):[N](#)
 - Clinical findings suggestive of central precocious puberty
 - Patient has been evaluated by pediatric endocrinologist.
- ☐ Preterm infant needing near-term equivalent (ie, 37 to 42 weeks' gestation) evaluation, as indicated by **1 or more** of the following(274)(275)(276):[N](#)
 - Birth at less than 32 weeks' gestation
 - Birth at less than 1500 g birth weight
 - Sickle cell disease, and results will impact treatment decision for stroke prophylaxis(278)[N](#)
- ☐ Stroke (ischemic) or transient ischemic attack, as indicated by **1 or more** of the following(250)(280)(281)(282)(283)(284):[N](#)
 - Aphasia(257)(300)
 - Ataxia(198)(300)
 - Decreased level of consciousness(300)
 - Dysarthria(257)
 - Follow-up of CT scan findings atypical for stroke(301)
 - Monocular blindness (ie, amaurosis fugax)(257)
 - Neonatal stroke, suspected (eg, altered mental status, encephalopathy, neurologic deficit, seizures)(289)(302)
 - Paresthesias(257)(300)
 - Transient global amnesia
 - Visual loss(113)(300)
 - Weakness or sensory loss on one side of body(257)
- ☐ Syncope, and associated symptoms or findings include **1 or more** of the following(303)(304)(305):[N](#)
 - Abnormality on neurologic examination
 - Bowel or bladder incontinence
 - Witnessed tonic-clonic seizure
- ☐ Trauma, as indicated by **1 or more** of the following(306)(307)(308)(309)(310):[N](#)
 - Carotid or vertebral artery dissection, suspected
 - Minor or subacute closed head injury (Glasgow coma scale score of 13 to 15) with cognitive or neurologic deficit, and CT scan contraindicated or not available, or results indeterminate(314)(315) [GCS Calculator](#)
 - Moderate or severe acute closed head injury (Glasgow coma scale score of less than 13), and CT scan contraindicated or not available, or results indeterminate [GCS Calculator](#)
 - Nonaccidental head trauma, suspected(312)(316)(317)
 - Subacute or chronic closed head injury with cognitive or neurologic deficit
- ☐ Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
 - Clinical need for repeat imaging, as indicated by **1 or more** of the following:
 - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
 - Need for interval reassessment that may impact treatment plan
 - Need for re-imaging either prior to or after performance of invasive procedure
 - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

Evidence Summary

Background

Some gadolinium-based contrast agents used for diagnostic and treatment guidance with MRI or MRA have been associated with nephrogenic systemic fibrosis in patients with chronic kidney disease or acute kidney injury.(4)(5)(6)(7)(8) **(EG 2)** Gadolinium deposition in the brain has been reported by multiple research groups; however, the clinical significance of this finding is unclear.(7)(8)(9)(10)(11) **(EG 2)** Specialty society guidelines recommend weighing the risks vs benefits when using gadolinium-based contrast agents for MRI. (12) **(EG 2)** MRI-based techniques may be contraindicated in patients with ferromagnetic intracranial aneurysm clips and certain types of ferromagnetic foreign bodies and cochlear implants.(13) **(EG 2)** Although these implants have traditionally included cardiac pacemakers and implantable cardioverter-defibrillators, multiple trials have validated the safe and effective use of pacemaker systems designed for MRI use, and a guideline supported by multiple specialty societies states that many patients with implantable cardiac devices, including those not originally intended for MRI scanning, can safely undergo MRI scanning under defined clinical indications. (14)(15)(16)(17) **(EG 1)**

Criteria

The evidence for the clinical indications found in this guideline includes 243 published peer reviewed articles, 30 specialty society or other evidence-based guidelines, 2 Cochrane systematic reviews, and 7 book sections.

For acoustic neuromas (vestibular schwannomas), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRI can characterize acoustic neuromas (vestibular schwannomas) with a high level of resolution, allowing for a more conservative approach to treatment using serial follow-up MRI to track tumor growth.(22) **(EG 2)** A retrospective chart review of 276 patients showed that 84% of the tumors most dangerous to patients (ie, those that grow most rapidly) were detected by the initial follow-up MRI at 6 months.(23) **(EG 2)**

For anatomy or structural defect evaluation, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Review article authors consider MRI to be the imaging modality of choice for malformations of the midbrain and hindbrain due to its ability to determine the location, degree, and extent of the malformation and identify associated supratentorial congenital anomalies. (25) **(EG 2)**

For anosmia, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Although CT scan may be useful for the evaluation of suspected obstructive causes of anosmia and visualization of nasal cavities, sinuses, and bony abnormalities, MRI can visualize the entire olfactory system as well as the temporal and frontal lobes; both tumors and vascular lesions of the temporal lobe can cause anosmia. (47) **(EG 2)** Anosmia is a common symptom of Parkinson disease and may precede motor symptoms by several years as part of a prodromal phase.(50)(51) **(EG 2)**

For cancer or neoplasm evaluation or staging, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Review articles and clinical guidelines state that MRI of the brain and spine is the gold standard for imaging of the central nervous system.(60)(61)(62)(63)(64) **(EG 2)** For pituitary or sellar tumor, an expert consensus guideline and a review article recommend MRI as first-line imaging for neuroendocrine disease of the pituitary or hypothalamus.(65)(66) **(EG 2)** For pituitary incidentaloma, a specialty society guideline recommends that patients with pituitary tumors incidentally noted on unrelated imaging (also known as pituitary incidentalomas) who have not undergone surgical treatment should be followed by MRI imaging at intervals determined by the size of the incidentaloma.(67) **(EG 2)**

For cerebral edema, including cytotoxic, vasogenic, interstitial, or combined etiologies, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Review articles indicate that MRI and CT scan offer complementary methods for evaluating underlying soft tissue lesion pathology.(137)(138) **(EG 2)**

For cerebrospinal fluid leak, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Brain MRI is recommended as part of the workup for suspected cerebrospinal fluid leak in patients with intracranial hypotension.(139)(140) **(EG 2)** Imaging findings suggestive of cerebrospinal fluid leak include pachymeningeal enhancement, engorged venous sinus, effacement of suprasellar and prepontine cisterns, subdural fluid collections, and decreased mamillopontine distance.(141) **(EG 2)**

For delirium, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRI is the imaging modality of choice for evaluating patients with altered mental status or suspected encephalopathy, although CT scan is often the initial investigation due to its availability and rapid performance.(143)(144) **(EG 2)** Specific toxic exposures, such as cyanide, ethylene glycol, methanol, and carbon monoxide, can cause characteristic MRI changes; imaging may be useful in managing patients with a complicated presentation.(145) **(EG 2)**

For dementia, known or suspected, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** For Alzheimer disease, a systematic review and meta-analysis of 33 studies (including 3935 participants with mild cognitive impairment) evaluated the diagnostic accuracy of structural MRI for early Alzheimer dementia and found that MRI cortical volume measurements had pooled sensitivity and specificity of 73% and 71%, respectively, for hippocampus and pooled sensitivity and specificity of 64% and 65%, respectively, for medial temporal lobe. The authors concluded that brain MRI was insufficient as a stand-alone diagnostic test for

Alzheimer disease; however, initial brain imaging was recommended to exclude other structural etiologies of dementia. Findings were limited by small study populations, interstudy heterogeneity, risk for bias, and low certainty of evidence; further larger, well-designed studies were recommended.(151) **(EG 1)** An expert consensus panel recommends performance of brain imaging once in every patient with suspected Alzheimer disease to exclude other structural etiologies of dementia and confirm specific imaging findings consistent with Alzheimer disease.(152) **(EG 2)** An open-label extension study of 2 clinical trials evaluating bapineuzumab for patients with mild to moderate Alzheimer disease further investigated the risk for the adverse effect of amyloid-related imaging abnormalities (ARIA), manifested as hemosiderin deposition (ARIA-H) or edema (ARIA-E), with potential for adjusted management. The investigators reviewed MRI results from 262 study participants and found that ARIA was associated with the presence of the APOE e4 alleles (hazard ratio 11.9 and 3.5 for 2 and 1 alleles, respectively), treatment with bapineuzumab therapy (hazard ratio 3.5), and the use of antithrombotic agents (hazard ratio 2.2). Findings were limited by potential for bias with retrospective analysis and variations in dosing and MRI protocols; further well-designed trials were recommended.(153) **(EG 2)** A health technology assessment concluded that brain imaging is most helpful in cases where the cause of dementia remains unclear despite prolonged follow-up (eg, 2 years or more); both CT scan and MRI can assist in detecting a vascular component of dementia.(154) **(EG 2)** An expert consensus panel recommends that a suspected diagnosis of vascular cognitive dementia be confirmed by evidence of significant cerebrovascular disease on either brain CT scan or MRI.(155) **(EG 2)** A consensus guideline states that brain MRI is preferable to brain CT to evaluate an individual with dementia if the presence of unsuspected cerebrovascular disease would change the clinical management; other indications for neuroimaging include duration of dementia less than 2 years, unexpected or unexplained cognitive or functional decline in a patient already diagnosed with dementia, recent significant head trauma, risk for intracranial bleeding, history of cancer with risk for brain metastases, or associated significant neurologic signs or symptoms (eg, gait abnormality, seizure, upper motor neuron signs).(156) **(EG 2)** A systematic review and meta-analysis indicates that CT scan is useful for exclusion of a potentially surgically treatable cause of dementia, such as tumor or subdural hematoma, as well as for the evaluation of the presence and extent of cerebrovascular disease.(157) **(EG 1)** However, MRI is more beneficial for identification of specific markers of dementing diseases, such as patterns of cerebral atrophy.(158)(159)(160)(161)(162) **(EG 2)** MRI is considered the preferred imaging modality for assistance with early diagnosis of dementia.(146)(163)(164) **(EG 2)** MRI may be useful to confirm dementia due to specific metabolic disorders, such as Wilson disease.(145)(165)(166) **(EG 2)** MRI is indicated before initiation of anti-amyloid monoclonal antibody therapy in patients with early Alzheimer disease to evaluate for findings associated higher risk of ARIA (eg, microhemorrhages), for surveillance for ARIA during anti-amyloid monoclonal antibody therapy, and to evaluate for stabilization or worsening of known ARIA.(148)(167) **(EG 2)**

For demyelinating disease (eg, multiple sclerosis), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** In patients presenting with a clinically isolated syndrome, such as unilateral optic neuritis, brainstem syndrome, or partial myelitis, diagnostic criteria for multiple sclerosis have been developed based on the demonstration of central nervous system lesions disseminated in space and disseminated in time on MRI imaging of the brain and, if appropriate, on spinal cord imaging.(177)(178) **(EG 2)** Requirements for radiographic dissemination in time criteria for multiple sclerosis include either the simultaneous presence of both gadolinium-enhancing (new) and nonenhancing (older) lesions on the same MRI, or new T2-weighted or gadolinium-enhancing lesions on a repeat MRI performed at any time after the initial onset of symptoms.(176)(177)(178) **(EG 2)** However, in some cases there is still significant inter-rater variability in assessing MRI results, and work continues to improve standardization of assessment of imaging findings, especially with regard to prognostic implications.(179)(180)(181)(182) **(EG 2)** A prospective cohort study of 1139 MRI scans in children and adolescents with acute central nervous system demyelination found that specific MRI parameters (eg, one or more T1-weighted hypointense lesions or one or more periventricular lesions) could be used to predict the diagnosis of multiple sclerosis.(183) **(EG 2)** Neuromyelitis optica spectrum disorder is often characterized by periependymal lesions located around the ventricular system or lesions involving the corticospinal tracts on brain MRI.(184) **(EG 2)**

For developmental delay, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A retrospective review of hospital records of 580 developmentally delayed children found that 58.6% of cases had an abnormal brain MRI; the authors concluded that the use of brain MRI may be effective for the diagnosis and management of developmental delay.(191) **(EG 2)** Review articles state that brain MRI may be useful in the evaluation of inborn errors of metabolism, which may present with neurologic symptoms that include developmental delay.(165)(194) **(EG 2)**

For dizziness or vertigo, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRI is a standard and widely accepted imaging modality for the evaluation of dizziness or vertigo in the setting of other potentially ominous signs or symptoms.(195)(196)(197)(198) **(EG 2)**

For epilepsy or seizure disorder, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Consensus guidelines recommend neuroimaging with either brain MRI or CT scan, preferably the former when it is available.(200)(201) (205) **(EG 2)** For children with epilepsy, an expert consensus guideline recommends neuroimaging with brain MRI for the following: status epilepticus; failure to control seizures, worsening seizures, or changes in seizure manifestations; seizures in children younger than 2 years, excluding those with febrile seizures; abnormal neurologic examination; a history of significant developmental delay, arrest, or regression; or a new-onset focal seizure.(200) **(EG 2)** A systematic review of studies using neuroimaging to identify patients with drug-resistant temporal lobe epilepsy who might benefit from focal cortical resection surgery found that the outcome was more favorable in patients with preoperative brain MRI demonstrating mesial temporal sclerosis; the authors concluded that brain MRI plays a pivotal role in evaluating surgical candidates with drug-resistant focal epilepsy of temporal lobe origin.(206) **(EG 1)** A systematic review of studies reporting on outcomes of epilepsy surgery found that an abnormal brain MRI, especially if it correlated with EEG findings, was more predictive of a good surgical result than a normal MRI.(207) **(EG 1)** A retrospective study of 177 children age 3 to 18 years

who presented with new-onset status epilepticus reported abnormal findings on brain CT scan or MRI in 36%; 8.5% had urgent or emergent pathology.(208) **(EG 2)**

For headache with a possible underlying structural cause, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Brain neuroimaging is indicated for those headaches associated with concerning signs or symptoms, including an increase in frequency or severity, description as the worst headache of the patient's life (eg, "thunderclap" headache), fever and meningeal signs, motor or sensory aura, and abnormal findings on neurologic examination.(215) **(EG 2)**

For hearing loss, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A specialty society guideline and review articles state that MRI is appropriate for the detection of acoustic neuroma (vestibular schwannoma) in appropriately selected patients, such as those with unilateral sensorineural hearing loss.(195)(196)(197) **(EG 1)**

For infection, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Brain neuroimaging is especially helpful in assessing potential brain abscesses, immunosuppression-related conditions, and viral or bacterial encephalitides.(223)(225)(228)(229)(230) **(EG 2)** When meningitis is suspected and the patient is at risk for an intracranial mass or midline shift, a head CT should be obtained before lumbar puncture is performed due to the risk of brain herniation.(225)(228) **(EG 2)** In uncomplicated acute meningitis, contrast-enhanced CT scan of the head may show leptomeningeal enhancement, though MRI of the brain tends to be more sensitive, particularly near the calvarium.(231) **(EG 2)**

For intracranial vasculitis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** MRI may demonstrate previous infarcts in different vascular distributions, suggestive of vasculitis. MRI is the most sensitive imaging modality for detecting the changes in cerebral vasculitis involving large, medium, and small vessels, with sensitivity estimated to be close to 100%; however, its specificity is much lower.(247)(248)(252) **(EG 2)**

For neonatal hypoxic-ischemic encephalopathy, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis of the prognostic value of tests for term neonates with hypoxic-ischemic encephalopathy found that brain MRI performed during the first week of life had a pooled sensitivity and specificity of 84% and 90%, respectively; diffusion-weighted MRI in the first week of life had a pooled specificity of 89% but a pooled sensitivity of only 58%.(254) **(EG 1)** MRI is the preferred imaging study for neonates with hypoxic-ischemic encephalopathy, and imaging abnormalities have been found to correlate with neurologic outcomes.(253) **(EG 2)**

For neurologic disease signs or symptoms, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Brain neuroimaging is indicated to help identify possible underlying causes when concerning neurologic signs or symptoms are present.(49)(250) **(EG 2)**

For Parkinson disease and other neurodegenerative conditions, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A meta-analysis of 10 studies with 595 patients found that the absence of dorsolateral nigral hyperintensity on iron-sensitive MRI sequences had sensitivity of 97.7% and specificity of 94.6% for differentiating Parkinson disease from healthy control patients.(264) **(EG 1)** MRI is helpful in differentiating Parkinson disease from atypical Parkinson disease and can provide additional evidence for the presence of other conditions, such as progressive supranuclear palsy.(261)(265)(266) **(EG 2)** MRI changes such as atrophy of the caudate nuclei may precede characteristic chorea in patients with Huntington disease by several years, facilitating a clinical diagnosis.(165)(267) **(EG 2)** Appropriate-use criteria state that brain MRI is an appropriate first test for suspected Parkinson or Huntington disease or dementia with Lewy bodies.(150)(268) **(EG 2)**

For precocious puberty (central), evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Brain MRI is indicated to rule out hypothalamic lesions such as hamartoma.(65)(272)(273) **(EG 2)**

For preterm infant needing near-term equivalent (ie, 37 to 42 weeks' gestation) evaluation, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review of 20 cohort studies with 1287 infants younger than 32 weeks' gestation, or less than 1500 g birth weight, evaluated 3 different findings (white matter abnormalities, brain abnormality, and diffuse excessive high signal intensity) noted on brain MRI performed at 37 to 42 weeks' gestation. Taken together, these findings had a pooled sensitivity of 77% and specificity of 79% for predicting cerebral palsy at age 18 months or older; for predicting motor dysfunction, brain MRI had a pooled sensitivity and specificity of 72% and 62%, respectively.(274) **(EG 1)** A prospective cohort study of 480 preterm infants (less than 28 weeks' gestational age) found that abnormalities seen on near-term (35 to 42 weeks' gestation) brain MRI or cranial ultrasound were associated with poorer neurodevelopmental outcomes at 18 to 22 months' corrected age; MRI with cerebellar abnormalities had a higher association with adverse outcomes. The authors concluded that neuroimaging at near-term equivalent for extremely preterm infants had value in predicting neurodevelopmental abnormalities at age 18 to 22 months.(275) **(EG 2)** A review article notes that brain MRI at term-equivalent age improves the prediction of later developmental disabilities; however, further research is needed.(277) **(EG 2)**

For sickle disease evaluation for stroke prophylaxis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A randomized controlled study of 196 children (mean age 10 years) with sickle cell disease, no history of stroke, and one or more silent cerebral infarcts on MRI compared the efficacy of standard care vs regular red blood cell transfusions to maintain a target hemoglobin concentration. At a median follow-up of 3 years, transfusion was associated with a lower incidence of stroke or new or enlarged silent infarct; the authors recommended that the detection of a silent cerebral infarct in a child with sickle cell anemia be followed by a discussion about treatment options.(279) **(EG 1)**

For stroke (ischemic) or transient ischemic attack, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** An expert consensus guideline recommends urgent brain MRI with diffusion-weighted imaging, preferably within 24 hours of symptom onset of stroke or transient ischemic attack.(285) **(EG 2)** Expert consensus guidelines and review articles recommend urgent brain imaging such as CT scan, or MRI if available and not contraindicated, for acute ischemic stroke prior to initiating any specific therapy, including thrombolytic medication.(250)(286)(287)(288) **(EG 2)** However, an evidence-based specialty society guideline recommends against routine brain MRI to exclude cerebral microbleeds before administration of thrombolytic medication; the guideline also notes that multimodal brain MRI should not delay thrombolytic administration.(280) **(EG 2)** An evidence-based national specialty society guideline recommends brain MRI with diffusion-weighted imaging for the diagnosis of ischemic stroke within 12 hours of symptom onset.(281) **(EG 2)** A consensus group recommends that neonates with suspected stroke be evaluated with MRI due to the lack of radiation.(289) **(EG 2)** A systematic review of 47 studies including 9078 patients with clinically diagnosed transient ischemic attack found that brain MRI with diffusion-weighted imaging demonstrated an acute ischemic lesion in only 34.3% of patients, with significant heterogeneity among studies; the authors concluded that MRI with diffusion-weighted MRI was not useful as a test to reclassify transient ischemic attack as stroke.(290) **(EG 1)** In a study of 215 consecutive patients with transient ischemic attack, CT scan revealed possible acute infarction in 7 of 154 patients, while MRI confirmed acute infarction in only 2 of these 7 patients. MRI also detected an acute infarct in 32.5% of the 154 patients with an initially normal baseline CT scan. The authors concluded that acute MRI, when available, is more advantageous than CT scan to investigate the etiology of lesions in patients with transient ischemic attack.(291) **(EG 2)** In reviews of the complementary roles of CT scan and MRI in acute stroke, the incremental benefit of CT scan over MRI included wider availability and more accurate detection of the presence of hemorrhage, while the incremental benefit of MRI over CT scan included greater sensitivity in detecting acute infarction and better ability to differentiate true stroke from stroke mimics.(292)(293)(294)(295)(296) **(EG 2)** Studies continue to assess the potential role of MRI for prognostication in the setting of stroke.(297)(298) **(EG 2)** Initial studies with MRI permeability imaging suggest, but have not confirmed, that MRI may predict hemorrhagic transformation after recanalization in the setting of acute ischemic stroke.(299) **(EG 2)**

For syncope, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Common etiologies include vasovagal response or orthostatic hypotension; MRI is nondiagnostic in most cases, unless syncope is associated with newly developed or focal seizures, focal neurologic findings, or episodes of syncope following head trauma.(303) **(EG 2)**

For trauma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Review articles indicate that CT scan is highly accurate for detecting injuries requiring emergency neurosurgery, such as hemorrhage or herniation; MRI is not usually indicated for the initial evaluation of traumatic brain injury because it takes longer to acquire, is generally less available, and requires additional safety precautions. However, MRI is more sensitive than CT scan to pathologic changes associated with mild traumatic brain injury and may also provide useful prognostic information.(307)(308)(309)(311) **(EG 2)** Appropriate-use criteria recommend brain MRI as the initial evaluation in nonemergent settings for any child with neurologic signs or symptoms, apnea, a complex skull fracture, other fractures, or injuries highly suspicious for nonaccidental head trauma.(312)(313) **(EG 2)**

Inconclusive or Non-Supportive Evidence

For bipolar disorder, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A meta-analysis of CT scan and MRI for this condition found robust evidence that changes in lateral ventricular size and white and gray matter characteristics were correlated with disease, but it concluded that substantial false-positive and false-negative errors remain, and further research is required.(18) **(EG 2)**

For psychosis without other neurologic findings (first episode), evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A retrospective review of 112 consecutive MRI and 204 consecutive CT scan studies in patients with a first episode of psychosis, but without other neurologic signs, found that 3% had brain lesions potentially contributing to psychosis on MRI, including brain tumor and HIV encephalopathy, and another 1.5% had brain lesions on CT scan; the authors indicated that routine MRI or CT scan imaging was unlikely to contribute to a significant change in management, but further prospective studies would be helpful.(19) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by

helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(318); 42 CFR 422.101(319)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(320); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(321); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(322)

Medicare Coverage Determinations: Medicare Coverage Database(323)

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Footnotes

[A] Non-central nervous system germ cell tumors that are associated with a high risk of central nervous system metastases include those associated with an elevated beta subunit of HCG level, nonpulmonary visceral metastases, extensive lung metastases, or neurologic symptoms, and nonseminomatous germ cell tumors associated with an elevated alpha fetoprotein level or predominance of choriocarcinoma.(78)(79)(82) [A in Context Link 1]

[B] Histiocytic and dendritic cell neoplasms are a heterogeneous group of rare hematologic disorders that arise from neoplastic proliferations of cells in the monocytic, histiocytic, or dendritic lineage that arise from common myeloid progenitors.(83)(84) They are classified on the basis of clinical, radiographic, histologic, and molecular features. Major categories include the plasmacytoid dendritic cell neoplasms; Langerhans cell and other dendritic cell neoplasms, including Langerhans cell histiocytosis; and histiocytic neoplasms, including Erdheim-Chester disease and Rosai-Dorfman disease.(83)(84)(85) Depending on the subtype, histiocytic and dendritic cell neoplasms may be characterized by central nervous system, orbital, dural, or skull involvement.(85)(86)(87) [B in Context Link 1]

[C] Macroincidentalomas are defined by size of 1 cm or greater.(67) [C in Context Link 1]

[D] Microincidentalomas are defined by size of less than 1 cm.(67) [D in Context Link 1]

Codes

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